

Quiz (Carolina Reaper)

Q1. Simplify $\sqrt{16}$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q2. Simplify $\sqrt[3]{27}$

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q3. Convert $\sqrt{x}$ to rational exponent form

- (A) $x^{1/2}$
- (B) $x^{1/3}$
- (C) $x^{2/3}$
- (D) $x^{3/4}$
- (E) $x^{4/5}$

Q4. Simplify $\sqrt{32}$

- (A) $4\sqrt{2}$
- (B) $6\sqrt{2}$
- (C) $8\sqrt{2}$
- (D) $4\sqrt{3}$
- (E) $4\sqrt{5}$

Q5. Simplify $\sqrt[3]{64}$

- (A) $2\sqrt[3]{2}$
- (B) $3\sqrt[3]{2}$
- (C) 4
- (D) $4\sqrt[3]{2}$
- (E) $4\sqrt[3]{3}$

Q6. Convert $x^{1/2}$ to radical form

- (A) $\sqrt[3]{x}$
- (B) $\sqrt{x}$
- (C) $x^{2/3}$
- (D) $x^{3/4}$
- (E) $x^{4/5}$

Q7. Simplify $\sqrt{48}$

- (A) $4\sqrt{3}$
- (B) $6\sqrt{2}$
- (C) $4\sqrt{2}$
- (D) $4\sqrt{5}$
- (E) $6\sqrt{3}$

Q8. Simplify $\sqrt[3]{128}$

- (A) $4\sqrt[3]{2}$
- (B) 5
- (C) $5\sqrt[3]{2}$
- (D) 4
- (E) $4\sqrt[3]{3}$

Open Ended Questions (FRQ)

Q9. Prove that $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$

Q10. Show that $x^{1/3} \cdot x^{1/3} = x^{2/3}$

Chef's Tips

- Remind students to rationalize denominators when simplifying radicals
- Encourage students to use properties of radicals to simplify expressions
- Caution students to check their work and verify their answers